

A Hybrid Correlation–PSO Feature Selection Framework with Extreme Learning Machines for High-Accuracy Cancer Classification in Microarray Data

1st Mr. Md. Anwar Ali
Department of Information Technology
MLR Institute of Technology
Hyderabad, India.
anwar@mlrinstitutions.ac.in

2nd Vaishnapallavi Devasani
Department of Information Technology
MLR Institute of Technology
Hyderabad, India.
vaishnapallavidevasani@gmail.com

3rd Soma Harsha vardhan
Department of Information Technology
MLR Institute of Technology
Hyderabad, India.
harshavardhan9103@gmail.com

4th Golla Reethu
Department of Information Technology
MLR Institute of Technology
Hyderabad, India.
reethugolla222@gmail.com

5th Gotrala Vishnu
Department of Information Technology
MLR Institute of Technology
Hyderabad, India.
vishnugotrala2117@gmail.com

Abstract—Cancer classification using microarray gene expression data remains challenging due to its extremely high dimensionality and the limited number of available samples, often leading to overfitting and poor generalization. Effective dimensionality reduction is therefore essential to identify a compact subset of informative genes that enhances diagnostic accuracy. This study presents, the combination of Correlation-based Feature Selection (CFS) and Particle Swarm Optimization (PSO) is presented as method to reduce redundant and irrelevant genes by selecting its informative subset effectively. The optimal subset of genes are subsequently classified using an Extreme Learning Machine (ELM), which is a rapid single-layer neural network suitable for high dimensional biomedical data. The experiments on benchmark multi-class microarray datasets (SRBCT, Lymphoma and MLL) show that the proposed CFS–PSO–ELM approach could reduce the feature space up to 92–98%, leading to an improved classification performance than classical J48, Random Forest and Genetic Algorithm classifiers. These results demonstrated the power of hybrid optimization and lightweight neural networks for precise cancer diagnosis

Keywords— *Cancer classification, microarray gene expression, feature selection, PSO, CFS, Extreme Learning Machine.*

I. INTRODUCTION

Cancer remains a common cause of deaths worldwide and early, accurate diagnose is important for the potential enhancement of patient survival. However, conventional diagnostic methods can be invasive, take time and at times be unreliable. The microarray gene expression which have tools that helps researchers to look how actually the thousands of genes working at same time. These datasets are very important for characterizing the cancer types, but the high dimensionality nature and the small size in terms of number of patients of the datasets makes the researchers more challenging. Actually, a microarray dataset includes thousands of genes but few samples. This imbalance is a

results in overfitting, poor performance and more computation in machine learning models. It is very important to pick out the most relevant genes, those which really different from one type of cancer to another. Feature selection can not only reduce the dimension but also increase the accuracy, accelerate computation and make the contribution of features to medical use more understandable.

Traditional feature selection methods, including filter-based and wrapper-based ones, have their own disadvantages. Filters are fast and wrappers achieve better accuracy but sometimes lags complex interdependencies among genes and computationally expensive. Hybrid models approach with the limitation of each method, which seek to combine benefits of both methods.

A hybrid method was employed, where CFS serves as a filter to eliminate irrelevant and redundant genes, while PSO acts as a wrapper to search the subset of features that result in highest classification performance. CFS rapidly trims down the dataset size and PSO further refines the feature subset by selecting from numerous combinations and collectively using swarm intelligence.

For classification, the Extreme Learning Machine (ELM), is a very fast and efficient neural network model. Unlike the conventional deep learning techniques, ELM takes almost no time to train as it does not require gradient descent based iterative weight updating. Both the speed and accuracy of ELM make the process well-suited for high-dimensional biological datasets.

Besides computational efficiency, the proposed CFS–PSO–ELM hybrid framework provides more practical benefits. By collapsing thousands of genes down into a compact, informative set, allows for cheaper genetic testing and clinical interpretation. The identified genes may be useful in cancer biology.

II. RELATED SURVEY

There is a growing interest in the hybrid and intelligent feature selection algorithms for cancer classification based on microarray data. Sun et al. (2019) proposed a hybrid algorithm is based on Relief-ACO model, which integrates a fast filter solution approach with a swarm-based wrapper search. Their method decreased dimension of 2 ~ 5% original genes and achieved better performance, while ACO needed high computation complexity and complicated parameter fine-tuning.[1]

Rostami et al. (2022) presented a multi-objective evolutionary strategy that took into account the minimizing of number of genes and maximizing classifier accuracy at the same time. Their approach provided Pareto-optimal solutions and produced better results for the Leukemia and Lymphoma datasets. Yet, this method was computationally intensive and results had no biological meaning.[2]

Tahmouresi et al. (2022) proposed the pyramid gravitational search algorithm that is a new meta-heuristic based on gravitational force. Compared with traditional PSO and GA, PGSA converged faster in the iteration process, but it had complexity as well as parameter sensitivity in the use of facilitation and adaptation.[3]

Several studies explored deep learning. Gupta et al. (2022) explored CNNs, autoencoders and DNNs for cancer classification and demonstrated that deep learning models can learn nonlinear interactions among genes. However, due to the small size of microarray data sets, they had a problem of overfitting and required a large amount of computational cost.[4]

Hybrid feature selection reviews, such as the work by Mohd Ali et al. (2022), emphasized the superiority of filter-wrapper combinations and metaheuristic-based selection. They highlighted PSO, GA, ACO and Firefly Algorithm as successful techniques. However these are time-consuming and lack biological verification.[5]

Rezaee et al. (2022) incorporated feature selection with deep neural networks, however the approach suffered from high variability associated with hyper-parameter sensitivity and lack of interpretability.[6] Similarly, Alharbi et al. (2023) which compared some ML models and has shown that the ensemble methods are efficient than single classifiers only for higher computational cost.[7]

Recent metaheuristic developments are the Binary Sand Cat Swarm Optimization by Pashaei et al. (2023) that was based on sand cat hunting style for the extraction of compact gene subsets. Its stochastic nature led to variations in different runs despite extremely frequent precision.[8] Isuwa et al. (2023), who tested a number of swarm algorithms and found that PSO had the best balance between accuracy and computational resources, although swarm methods were dominated in this respect.[9]

Newer optimizers such as SKF-M (Ahmad Zamri et al., 2023) and Relief-NRO (El-Mageed et al., 2024) provided better convergence, stability but they also consumed computational resources and had weak biological significance[10][14]. Alromema et al. (2023) and Yang et al. (2024) proposed advanced hybrid pipelines with filter + metaheuristic approaches and an enhanced PSO variant, which independently showed a greater performance in terms of accuracy and smaller gene subsets at a higher algorithmic complexity. [11][12] Finally, Nagra et al. (2024) extended PSO using adaptive weights with an universal testing in different classifiers yet once again faced the performance vs. complexity trade-off.[13]

III. PROPOSED MODEL

The proposed architecture gives a complete flow process for classifying cancer based on microarray gene expression data. The workflow starts with a raw microarray dataset that contains gene expressions with thousands of genes per patient. Since such datasets generally have missing entries and inconsistencies, Data Pre-Processing is the first process, which is aimed at cleaning the data, filling the missing values, and normalizing (Fig.1).

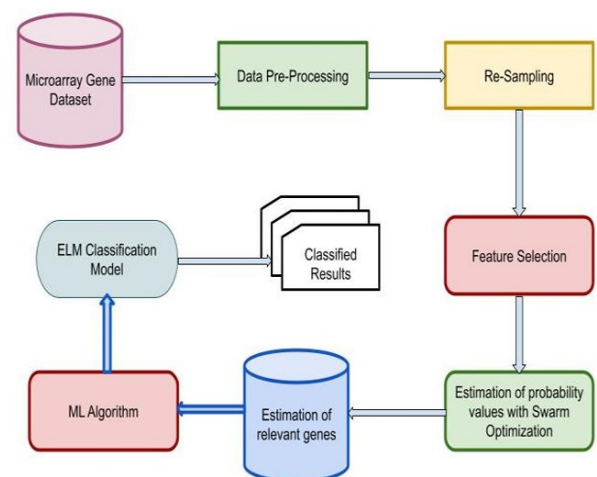


Figure 1 shows the architecture for the proposed model

Next, the re-sampling takes the processed data. This makes sure that the size of the classes is the same. Additionally, most microarray datasets have imbalanced classes because some cancers may occur less frequently than others. Therefore, re-sampling ensures that all classes have the same weight. The algorithm is prevented from giving more importance to the most common cancers.

Feature selection is a very important step. There are so many genes in microarray data that selecting the helpful ones is very crucial. Correlation scoring could be a similar method for discarding redundant features. Nevertheless, filter methods may fail to find the best combinations. That is when PSO decides to try lots of different gene combinations.

After that, the extreme learning machine (ELM) wraps up the procedure for classification. ELM's modules features quick working memory since it silently ignores slow changes.

Because of that, it manages quite complex biologically based input data.

The last stage uses an Extreme Learning Machine (ELM) for prediction. It is a fast working memory because it quietly leaves out slow changes. Therefore, it handles quite complex biologically based input data well. With the best set of genes found, ELM is able to correctly classify all the samples into their respective cancer classes.

A) Dataset Collection: The datasets are open-source gene expression microarray data, such as Lymphoma, MLL, and SRBCT, that have highly complicated genetic traces of patients. All datasets have gene expressions for thousands of genes for each patient, the range being from 2,000 to almost 12,600, but still, the number of patients is less than 200, the feature selection with thousands of variables and very few samples. The datasets are sourced from Kaggle and Open ML.

Each gene is an individual tissue sample, all the vertical slices showing genes and the tag being the type of cancer. Because there are too many genes involved, it uses CFS to remove the ones that are not needed. After reducing the list, PSO changes it in such a way that only the most influential genes with high signal-carrying abilities affect the output Table.1 .

Gene_1	Gene_2	Gene_3	Gene_4	...Gene_15	Label
0 5.221	8.134	1.238	4.556	7.441	Lymphoma
1 4.981	9.014	1.112	4.334	6.780	MLL
2 6.441	7.901	1.300	4.998	7.122	SRBCT

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As Table 1 illustrates overview of Gene Expression dataset for Cancer Classification

B) Pre-Processing: Preparing gene data is a important in making the results more accurate. There is information for thousands of genes, but the number of samples is quite small - therefore by sorting, adjusting, or correcting imbalances it becomes easier to select the most important features as well as to train models properly.

a)Data Cleaning: Fixing the data means removing the data such as duplicates. Instead of removing the data ,methods like K-NN estimates the missing numbers. Unnecessary parts and noise are removed so that the data remains clean. In this way, data becomes complete, trustworthy, and ready for use.

b)Normalization: Normalization Gene expression levels vary significantly from one gene to another. To prevent high-value genes from dominating the results during training the data is scaled, usually by means of Z-score. This scaling brings all features to a similar range. As a result of this, PSO can be used more efficiently with ELM. The whole training is not only more stable but also has higher accuracy.

c) Resampling: Microarray data is often characterized by unbalanced class sizes, i.e., certain cancer types have fewer cases than others. Therefore, rather than just accepting this

fact, a method called SMOTE is employed to balance the data. SMOTE generates new synthetic instances for the minority classes by combining real ones. Thus, models are not overfitted to common types and at the same time, they do not underperform on the rarer ones.

e)Feature Selection: By employing the technique, CFS adjustment alters the ones with which it has recognized search behaviour assistance in order to be able to locate the optimal feature combination for data classification. Every particle represents a specific gene set, and PSO is trying to find the gene combination that yields the best ELM performance.

C) Training &Testing: Data splitting is done 80% for training and 20% for testing. The training is employed here to help the CFS-PSO-ELM model takes key gene cancer features by learning from the gives examples instead of making random guesses. The rest 20% is for separate test, a performance evaluation, the prevention mechanism of overfitting, and it also gives a hint to the model's ability of handling new data.

D) Algorithms:

1)Correlation based FeatureSelection(CFS): CFS works like an intelligent filter that picks out genes which have a strong association with the cancer type but not genes that are very similar to other ones. It avoids repetitions of info while data gets smaller but remains of the highest quality thus the following stages can be dealt with more comfortably due to the quicker setup.Reducing data to the minimum necessary helps focus on discovering patterns.

$$\text{Merit}_s = \frac{k \cdot \overline{r}_{cf}}{\sqrt{k + k(k-1) \overline{r}_{ff}}} \quad (1)$$

2)Particle Swarm Optimization (PSO): PSO method modifies the CFS-recognized features by means of search behavior in order to find the best feature combination for data classification. A particle corresponds to a certain gene set, and PSO selects the one that yields the best ELM results.

$$v_i(t+1) = wv_i(t) + c_1r_1(pbest_i - x_i(t)) + c_2r_2(gbest - x_i(t)) \quad (2)$$

$$x_i(t+1) = \sigma(v_i(t+1)) \quad (3)$$

3)Extreme Learning Machine (ELM): ELM is the final classification model used in this study. It is a single-hidden-layer neural network, and there is no need for iterative training. Weights for input and biases are randomly chosen, whereas outputs are calculated analytically.

$$H = \varphi(XW + b) \quad (4)$$

$$\beta = (H^{\{T\}}H + \lambda I)^{-1}H^{\{T\}}T \quad (5)$$

IV.RESULTS AND DISCUSSION

The hybrid approach was applied to gene data sets like Lymphoma, MLL, and Breast Cancer. In order to verify the strength, researchers compared CFS, PSO, and ELM with traditional methods. Neural Networks(ELM) worked together without depending on the old Models that slow down the process of complex medical information. Results showed this handles the Logical data more effectively than usual techniques.

A)Accuracy Comparisons:

Table 2. Accuracy Comparison of Classification Methods

Method	Accuracy (↑)
J48 Decision Tree	85.62
Random Forest	89.23
Genetic Algorithm (GA)	90.51
Proposed CFS-PSO-ELM	97.85

Accuracy values obtained for all four classifiers are given in Table 2.The proposed Hybrid CFS-PSO-ELM algorithm achieves beating older methods like J48 Decision Tree at 85.62%,89.23%, or Genetic Algorithm picking features with 90.51%.Figure 2 illustrates how the hybrid system learns with experience and accuracy gets stable step by step. At the same time, both training and validation losses decline nicely, no jumping and no wild changes, instead of memorizing noise, learning patterns properly.

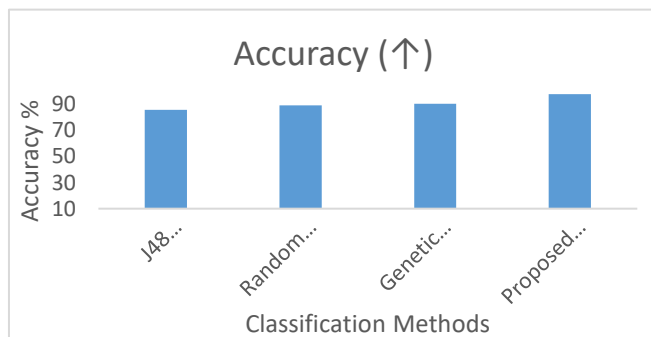


Figure 2 shows the accuracy graph

B)Precision Comparisons:

Table 3. Precision Comparison of Classification Methods

Method	Precision (↑)
J48 Decision Tree	83.7
Random Forest	88.42
Genetic Algorithm (GA)	89.12
Proposed CFS-PSO-ELM	97.45

Accuracy for each classifier in Table 3 – The proposed Hybrid CFS-PSO-ELM reaches 97.45%, outperforming older approaches like J48 Decision Tree at 83.7% or Random

forest sitting near 88.42%. The method based on Genetic Algorithm, which lands around 89.12%.It shows that this combination works far better with respect to cutting. wrong investigations-a point not to be taken lightly because health-related mistakes end up being quite costly. On the other hand, detection could indicate illness (Fig.3).

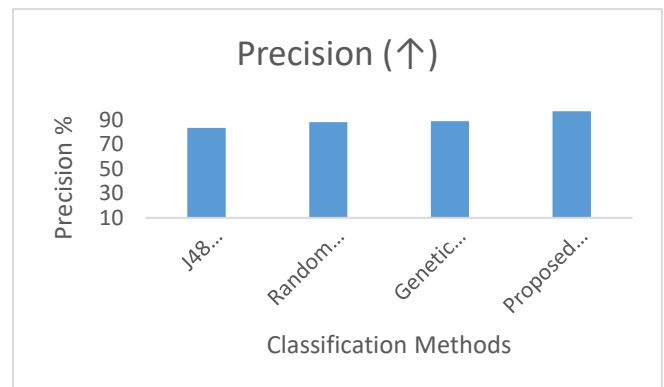


Figure 3 shows the precision graph

C) Recall Comparisons:

Table 4. Recall Comparison of Classification Methods

Method	Recall (↑)
J48 Decision Tree	84.12
Random Forest	87.5
Genetic algorithm (GA)	89
Proposed CFS-PSO-ELM	97.2

The value of recall for the four models can be seen in Table 4.The proposed Hybrid CFS-PSO-ELM algorithm achieves an accuracy rate higher than J48 Decision Tree with accuracy of 84.12%. Model outperforms Random Forest, which manages 87.5%. On top of that, it does more than the 89% achieved by the Genetic Algorithm. That jump demonstrate that this model can identify real cancer instances more effectively. In healthcare settings, identifying every actual case is important because missing even a single one would be potentially damaging (Fig.4).

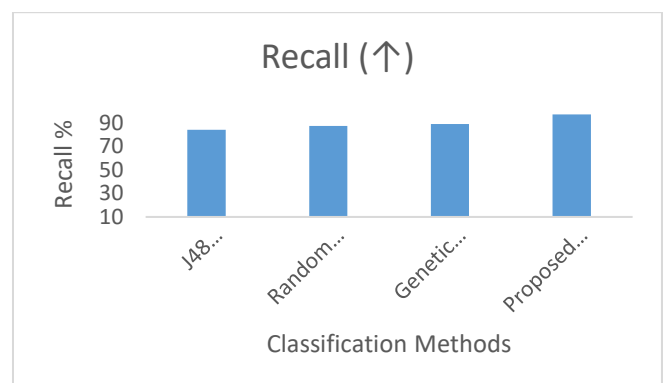


Figure 4 shows the recall graph

D) F1-Score Comparisons:

Table 5. F1-Score Comparison of Classification Methods

Method	F1-Score (↑)
J48 Decision Tree	83.9
Random Forest	87.9
Genetic Algorithm (GA)	89.05
Proposed CFS-PSO-ELM	97.32

Table 5 contrasts the F1-score, which can be seen as the signal that achieving the suggested Hybrid CFS-PSO-ELM architecture with 97.32%, the greatest F1-Score far exceeding that of the J48 Decision Tree performed at 83.9%; Random Forest at 87.9%; Genetic Algorithm (GA) at 89.05%. Given that the F1-score is the harmonic mean of precision and recall, This confirms that the hybrid approach maintains an improvement. Perfect equilibrium between accurately diagnosed cancer cases and reduce inaccuracy in predictions (Fig.5).

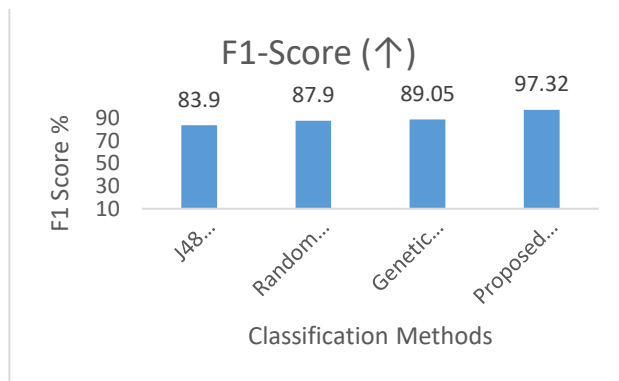


Figure 5 shows the F1-Score graph

V.CONCLUSION AND FUTURE SCOPE

In this investigation, several methods were employed: ELM for gene classification on microarrays are executed using correlation filters, clever swarm search, and other rapid neural learning addresses the challenge posed by big data size. Sources includes sample and genes. The CFS component removed unwanted features. however, enhanced the overall set of genes and precision by means of decreased overlaps. The arrangement for adding issues used an ELM classifier, noted for its rapid convergence and consistent actual performance which is shown by data on breast cancer and MLL. The results are of 97.85% right calls accuracy, good precision, recall, and F1 compared with J48, Random Forest, or GA. The CFS-PSO-ELM looks to be follows scales gracefully and deals with heavy delayed medical information helps to reduce unnecessary features and keeping it sharp so it could be useful for tumor detection, discovering bio-signals, resolving biogenetic data. Better trait filtering depends on the study, which offers a great basis. The CNNs variants could be fed autoencoders and graph nets to detect on complex genes illustrate how old concepts are no longer followed. A deep feature CFS-PSO can guide the model in discriminating among kinds. Methods like NSGA-II or MOPSO will eventually be explored aimed at reducing feature number despite still reaching a high degree of precision for clearer sight, results instead of black box reasoning. There are some

genes that are so important, and methods like SHAP or LIME aid in medical diagnosis. It would be great to experiment with this on other databases as well as with examples include Array Express, The Cancer Genome Atlas, and GEO. Once implemented via cloud configurations or tiny modular services might be given CFS-PSO-ELM set. Doctors and scientists will find gene analysis very useful.

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